INCH-POUND

MIL-C-70996 (AR)
AMENDMENT 4
7 April 2000
SUPERSEDING
AMENDMENT 3
19 September 1997

MILITARY SPECIFICATION

CARTRIDGE, HE, 81MM, M821E1/M889E1 W/FUZE M734/M935
LOADING, ASSEMBLING AND PACKING

MIL-C-70996 was inactivated after 13 January 1997 for new design

This amendment forms a part of MIL-C-70996(AR), dated 9 May 1991, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete: “M821E1”, and substitute: “M821A1” wherever it appears in the specification.

Delete: “M889E1”, and substitute: “M889A1” wherever it appears in the specification.

PAGE 2

2.1.2, DRAWINGS, Add the following after ARDEC DRAWINGS:

“Earle Naval Weapons Station, Code 5021, Colts Neck, NJ 20632

PACKAGING DRAWINGS

6213320 - Packing and Marking for Container, Ammo, PA156 and PA157 for Cartridge 81mm”.

AMSC N/A FSC 1315
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3.3: Delete in its entirety and substitute:

"3.3 Surface cavitation. Cavities formed in Segment D (see 3.7) due to chipped or broken explosive from the sidewall of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum dimension of any cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the fuze booster would rest. Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in the computation. Disregard any cavitation in the fuze well above Segment D."

* 3.5: Delete in its entirety and substitute:

"3.5 High explosive on threads. Threads shall be free of high explosive as described in the applicable drawing."

3.7: Delete in its entirety and substitute:

"3.7 Cavitation in high explosive charge.

3.7.1 Projected areas and dimensions. Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table III when determined as specified in 4.4.3.3 and 4.5.3. Cavities having a maximum projected length of 1/32 inch or less shall be disregarded, provided they are separated by at least 1/2 inch of solid cast."

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Table III: Delete in its entirety and substitute:

"TABLE III. Cavitation requirements.

	SEGMENTS (see Figure 1)			
Sum of projected area of the cavities, excluding pipes, cracks and annular rings, square inch	1/16	1/4	1/2	1/2
Projected length of any cavity excluding pipes, cracks and annular rings, inch	1/4	1/2	1/2	3/4
Piping cavities, maximum projected area, square inch	0	1/4	1/2	NA
Piping cavities, maximum projected width, inch	0	1/2	1/2	NA
Cracks, maximum projected width, inch	1/20	1/20	1/20	NA
Annular rings, maximum projected width, inch	0	0	1/4	NA"

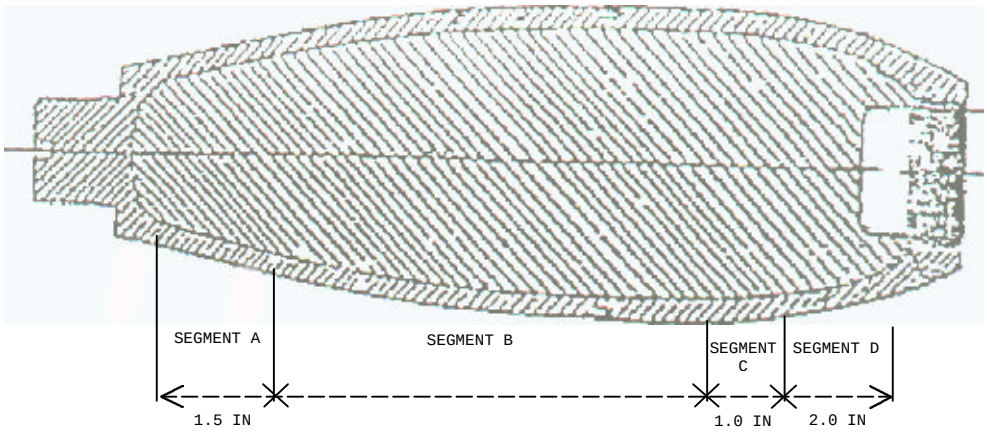


FIGURE 1. SEGMENTAL DEFINITION

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PAGE 4

3.7.2: Delete in its entirety and substitute:

"3.7.2 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon the cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered."

3.7.3 Delete in its entirety and substitute:

"3.7.3 Cracked charges. Not more than one (1) transverse crack shall be permitted in Segment A and not more than a total of three (3) transverse cracks shall be permitted in Segments A, B and C, combined. Cracks shall be disregarded in Segment D."

3.7.4 Delete in its entirety and substitute:

"3.7.4 Annular rings. The presence of annular rings 1/16 inch, maximum, and extending from the shell wall 1/4 inch into the charge shall be disregarded."

3.7.5 Delete in its entirety and substitute:

"3.7.5 Cavities extending into two (2) segments. The entire area of cavities extending into two (2) segments shall meet the requirements of both segments."

3.7.6 Delete in its entirety and substitute:

"3.7.6 Base separation. The separation between the base of the projectile and the explosive charge shall not be greater than 0.015 inch."

PAGE 11

* 4.4.2.2: Add Critical 1 as follows:

"1... High explosive on threads ... 100% ... 3.5 ... Gage/visual".

4.4.2.2: Add Special B as follows:

"B ... Radiographic inspection ... 4.4.3.3 ... 3.7 ... 4.5.3".

* 4.4.2.2, Minor 202: Delete in its entirety.

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4.4.2.3, Special A: Delete “Gage” and substitute “Visual”.

4.4.2.3, Page 1 of Amendment 1: Delete “4.4.2.3” and substitute “4.4.2.2”.

PAGE 12

* 4.4.2.3, Under Conformance Criteria for Critical 1, Major 101 and Major 102, delete “100%” and substitute “100% (a)”.

* Add Note (a) as follows:

"(a) If LAP of the propelling charge and the cartridge are performed at the same installation, this paragraph is not required".

PAGE 13

4.4.2.4, Critical 1: Delete in its entirety and substitute:

"Major 111 – Ignition cartridge not properly assembled or fully seated".

PAGE 14

4.4.2.4, Major 102: Delete in its entirety.

* Major 105, 106 and 107: Delete in their entirety and substitute:

"105 Fin assembly fails to meet minimum assembly torque (while cement still wet) ... Level III ... 3.2 ... Test.

106 Fuze assembly fails to meet minimum assembly torque (while cement still wet) ... Level III ... 3.2 ... Test.

107 Ignition cartridge fails to meet minimum assembly torque (while cement still wet) ... Level III ... 3.2 ... Test.”

* Add Majors 111, 112 and 113 as follows:

"111 Obturator pull test ... Level III ... 3.2 ... 4.4.3.2

112 Obturating band pull test ... See 4.4.3.2 ... 3.2 ... Gage/4.5.2

113 Obturating band not welded ... 100% ... 3.2 ... Visual”

* Minor 201, Conformance criteria delete “Level III” and substitute “Level V”.

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4.4.2.5: Add “/6213320” after “12630601”.

PAGE 16

4.4.2.6: Add “/6213320” after “12630601”.

* Add Majors 104 and 105 as follows:

"104 Metal ends loose or distorted ... Level III ... 5.1 ... Visual/manual

105 Pull tab missing or improper ... Level III ... 5.1 ... Visual”.

PAGE 17

4.4.2.7: Delete “12630600” and substitute: “12630601”.

PAGE 17a

4.4.2.8: Add the following:

CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.4.2.8	TITLE Packing and Marking for Container, Ammo, PA156 & PA157 for Ctg 81mm (Prior to Closing)	SHEET 1 OF 1		DRAWING NUMBER 6213320
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Support assembly damaged, missing or improperly placed	Level III	5.2	Visual
102	Containers loose or improperly packed in metal can	Level III	5.2	Manual/Visual
Minor				
201	Circular firing scale not present in every third metal box	Level V	5.2	Visual
202	Plastic sheet not present in every six boxes	Level V	5.2	Visual
203	Evidence of poor workmanship	Level V	5.2	Visual

* 4.4.2.8, Major 102, delete in its entirety and substitute:

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"102 Containers improperly packed in metal can ... Level III ... 5.2 ... Manual/Visual".

* Minor 201, under examination or test, change the word "circular" to "graphical" and the word "box" to "container".

* Minor 202, under examination or test, change the word "boxes" to "containers".

PAGE 17b

4.4.2.9: Add the following:

CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.4.2.9	TITLE Packing and Marking for Container, Ammo, PA156 & PA157 for Ctg 81mm (After Closing	SHEET 1 OF 1		DRAWING NUMBER 6213320
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Contents of box loose	Level III	5.2	Manual
102	Cover assembly not secure	Level III	5.2	Manual
Minor				
201	Marking missing, misleading or unidentifiable	Level V	5.3	Visual
202	Evidence of poor workmanship	Level V	5.2	Visual
NOTES:				

* 4.4.2.9, Major 101, under examination or test, change the word "box" to "container".

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4.4.2.10: Add the following:

CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.4.2.10	TITLE Support, Propellant Charge	SHEET 1 OF 1		DRAWING NUMBER 12977198
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 12630535/ 12630672 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width of opening	(a)	3.2	Gage
102	Inside length	(a	3.2	Gage
103	Radius to outside surface of rib (3 places)	(a)	3.2	Gage
104	Inside radius to support wall	(a	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES:
(a) Sample (3) parts from each mold cavity at the start of each mold run. All dimensions shall be as specified in the drawing prior to production run.

PAGE 18

* 4.4.3.2, Delete paragraph in its entirety and substitute:

"4.4.3.2 Obturating band pull test. This test shall be performed at the start of a production run and after each 100 cartridges that are welded during that run. The run shall be continuous for the entire time. If the welding run is stopped for any reason, a new test sample must be tested and the test sampling process above shall resume."

PAGE 19

4.4.3.3: Delete in its entirety and substitute:

"4.4.3.3 Cavitation (see 3.7). The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235, using CSP-1 plan, having the following parameters:

i = 1470
f = 1/10
S = 4120

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Any cartridge having cavitation in excess of that permitted by the applicable requirements for any segment shall be classed defective. The test shall be performed as specified in 4.5.3 with equipment in accordance with 4.4.4."

PAGE 20

* 4.5.2, Delete paragraph in its entirety and substitute:

"4.5.2 Obturating band pull test. Properly install and weld an obturating band onto a simulated 81mm body groove of the correct size and shape (see projectile body drawing) with a relief for a pull test hook. Apply the required load perpendicular (radially and in a direction away from the center axis) to the weld. If the weld fails below the minimum weld strength requirement, replace the band on all rounds since the last test. If the weld fails above the maximum weld strength requirement, replace the band on all rounds since the last test."

PAGE 22

5.2.1: Add "/6213320" after "12630601".

5.3.1: Add "/6213320" after "12630601".

The margins of this amendment are marked with an asterisk, or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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